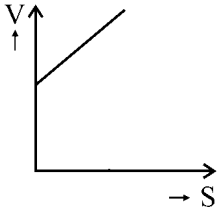
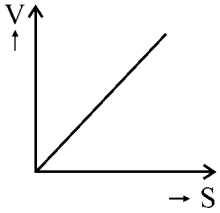
PHYSICS

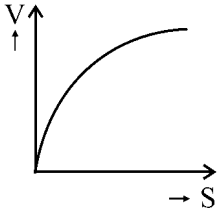
1) In an experiment, the period of oscillation of a simple pendulum was observed to be 2.63 s, 2.56 s, 2.42 s, 2.71 s and 2.80 s. The mean absolute error is :-

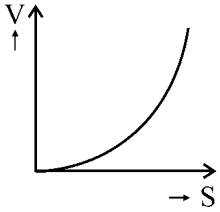
(1) 0.11 s
(2) 0.12 s
(3) 0.13 s
(4) 0.14 s

2) A body starting from rest moves along a straight line with a constant acceleration. The variation of speed (V) with distance (S) is represented by the graph :-

(1)


(2)


(3)


(4)


3) If a ladder weighing 250 N is placed against a smooth vertical wall having coefficient of friction between it and floor is 0.3. Then what is the maximum force of friction available at the point of contact between the ladder and the floor ?

(1) 75 N
(2) 50 N
(3) 35 N
(4) 25 N

4) If work done in stretching a spring from x_0 to $2x_0$ is W, then work done in stretching this spring from $2x_0$ to $3x_0$ will be :-

(1) W

(2) 5W

(3) $\frac{3W}{5}$

(4) $\frac{5W}{3}$

5) A torque of 2 N–m produces an Angular acceleration of 2 rad/sec^2 in a body. If its radius of gyration is 2m. Its mass will be :-

(1) 2 kg

(2) 4 kg

(3) 1/2 kg

(4) 1/4 kg

6) If temperature of ideal black body increased by 10% then percentage increase in quantity of radiation emitted from its surface will be :-

(1) 10%

(2) 40%

(3) 46%

(4) 100%

7) When the temperature of an ideal gas is increased by 600 K, the velocity of sound in the gas becomes $\sqrt{3}$ times the initial velocity in it. The initial temperature of the gas is

(1) –73°C

(2) 27°C

(3) 127°C

(4) 327°C

8) Two particles A and B of equal masses are suspended from two massless springs of spring constants k_1 and k_2, respectively. If maximum velocities during oscillation are equal the ratio of amplitudes of A and B is :-

(1) $\frac{k_1}{k_2}$

(2) $\frac{k_2}{k_1}$

(3) $\sqrt{\frac{k_1}{k_2}}$

(4) $\sqrt{\frac{k_2}{k_1}}$

9) Two similar sonometer wires given fundamental frequencies of 500 Hz. These have same tensions. By what amount the tension be increased in one wire so that the two wires produce 5 beats/sec.

(1) 1%

(2) 2%

(3) 3%

(4) 4%

10) A body weighs 700 gm wt. on the surface of the earth. How much will it weigh on the surface of a planet whose mass is $\frac{1}{7}$ and radius half of that of the earth-

(1) 200 gm wt

(2) 400 gm wt

(3) 50 gm wt

(4) 300 gm wt

11) The liquids A, B and C are allowed to flow in identical capillaries identical conditions, but the volumes of liquids collected in equal time interval are $2cm^3$, 4 cm^3 and 8 cm^3. The ratio coefficients of viscosity of liquids A, B and C is :-

(1) 1 : 2 : 4

(2) 1 : 1 : 1

(3) 1 : 4 : 16

(4) 4 : 2 : 1

12) A substance of mass m kg requires a power input of P watt to remain in the molten state at its melting point. When the power is turned off, the substance completely solidifies in time t sec. The latent heat of fusion of the substance is :-

(1) $\frac{Pm}{t}$

(2) $\frac{Pt}{m}$

(3) $\frac{m}{Pt}$

(4) $\frac{t}{Pm}$

13) A monkey of mass 20 kg rides on a 40 kg trolley moving at a constant speed of 8 m/s along a horizontal track. Frictional force can be neglected. If the monkey jumps vertically, with respect to the moving trolley, the catch the overhanging branch of tree, the speed of the trolley after the monkey has jumped off is :-

(1) 16m/s

(2) 12m/s

(3) 8m/s

(4) 6m/s

14) A thin uniform circular disc of mass M and radius R is rotating in a horizontal plane about an axis passing through its centre and perpendicular to the plane with angular velocity ω. Another disc of same mass but half the radius is gently placed over it coaxially. The angular speed of the composite disc will be :-

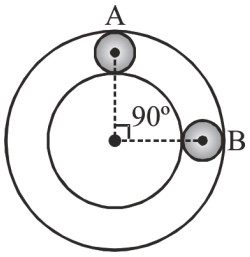
(1) $\frac{5}{4}\omega$

(2) $\frac{4}{5}\omega$

(3) $\frac{2}{5}\omega$

(4) $\frac{5}{2}\omega$

15) Body A of mass m and B of mass 3m move towards each other with velocities V and 2V respectively from the positions as shown, along a smooth horizontal circular track of radius r. After



the first elastic collision, they will collide again after the time :-

(1) $\frac{2\pi r}{V}$

(2) $\frac{\pi r}{2V}$

(3) $\frac{\pi r}{V}$

(4) $\frac{2\pi r}{3V}$

16) If a particle moves in a circle describing equal angles in equal times, its velocity vector :-

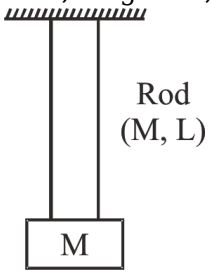
(1) Remains constant

(2) Changes in magnitude

(3) Changes in direction

(4) Changes both in magnitude and direction

17) A block of mass 'M' is suspended by a rod of mass 'M', length 'L', area of cross-section 'A' and



Young's modulus 'Y'. Find increment in length of rod.

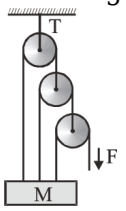
(1) $\frac{3MgL}{2AY}$

(2) $\frac{MgL}{2AY}$

(3) $\frac{2MgL}{AY}$

(4) $\frac{MgL}{AY}$

18) Consider the pulley system in the diagram below. The unknown force F being applied is just sufficient to hold the system in equilibrium. The block has mass M, while the pulleys and ropes have negligibly small masses. What is the tension T in the upper cable (i.e. the cable connecting the top pulley to the ceiling) in terms of M and the acceleration due to gravity g only.



(1) $\frac{8Mg}{7}$

(2) Mg

(3) $\frac{Mg}{5}$

(4) $\frac{8Mg}{5}$

19) A tuning fork produces four beats per second with two open organ pipe having length 30 cm and 31 cm. Find the frequency of T.F.

(1) 120 Hz

(2) 124 Hz

(3) 240 Hz

(4) 244 Hz

20) A particle of charge -16×10^{-18} C moving with velocity $10ms^{-1}$ along the x-axis enters in a region where a magnetic field of induction B is along the y-axis and an electric field of magnitude 10^4 V/m is along the negative z-axis. If the charged particle continues moving along the x-axis, the magnitude of B is :-

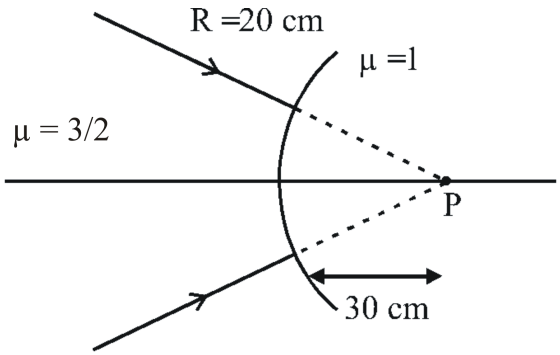
(1) 10^{-3} Wb/m^2

(2) 10^3 Wb/m^2

(3) 10^5 Wb/m^2

(4) 10^{16} Wb/m^2

21) The image for the converging beam after refraction through the curved surface is formed at :-



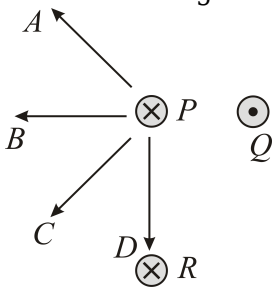
(1) 40 cm

(2) 40/3 cm

(3) –40/3 cm

(4) 20 cm

22) In figure shows three long straight wires P, Q and R carrying currents normal to the plane of the paper. All three currents have the same magnitude. Which arrow best shows the direction of the resultant force on the wire P



(1) A

(2) B

(3) C

(4) D

23) An ideal coil of L = 10 H is joined in series with a resistance of 5Ω and battery of 5 volt. 2 seconds after joining, the current flowing in the circuit (in ampere) will be :-

(1) e^{-1}

(2) $1- e^{-1}$

(3) 1 – e

(4) e

24) A coil of area 80 square cm and 50 turns is rotating with 2000 revolutions per minute about an axis perpendicular to a magnetic field of 0.05 Tesla. The maximum value of the e.m.f. developed in it is

(1) 200 π volt

(2) $\frac{10\pi}{3}$volt

(3) $\frac{4\pi}{3}$volt

(4) $\frac{2}{3}$volt

25) Consider two identical cells each of emf E and internal resistance r connected to a load resistance R.

Column-I		**Column-II**	
(i)	Maximum power transfer to load if cells are connected in series	(p)	$\frac{E^2}{4r}$
(ii)	Maximum power transfer to load if cells are connected in parallel	(q)	$\frac{E^2}{2r}$
(iii)	For series combination of cells	(r)	$E_{eq} = E, r_{eq} = \frac{r}{2}$
(iv)	For parallel connection of cells	(s)	$E_{eq} = 2E, r_{eq} = 2r$

Now, match the given columns and select the correct option from the codes given below :
Code :

(1) i-p, ii-q, iii-p, iv-s

(2) i-r, ii-p, iii-q, iv-s

(3) i-q, ii-q, iii-s, iv-r

(4) i-p, ii-p, iii-q, iv-s

26) What should be the angle between two plane mirrors so that whatever be the angle of incidence, the incident ray and the reflected ray from the two mirrors be antiparallel to each other :-

(1) 60°

(2) 90°

(3) 120°

(4) 175°

27) The magnetic field of given length of wire for single turn coil at its centre is B then its value for three turns for the same wire is :

(1) $\frac{B}{9}$

(2) 9B

(3) $\frac{B}{3}$

(4) 3B

28) The magnetic field in a plane electromagnetic wave is given by :
$B_y = 2 \times 10^{-7} \sin(\pi \times 10^3 x + 3\pi \times 10^{11} t)T$
Calculate the wavelength.

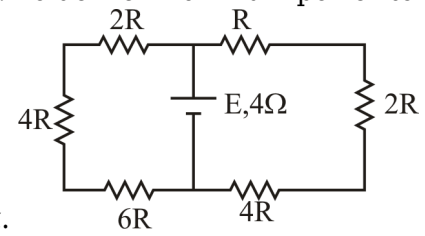
(1) $\pi \times 10^3$ m

(2) 2×10^{-3} m

(3) 2×10^{3} m

(4) $\pi \times 10^{-3}$ m

29) A battery of internal resistance 4Ω is connected to the network of the resistance as shown in the figure. To deliver maximum power to the network the magnitude of resistance R in Ω should be $\frac{x}{21}$.



Find *x*.

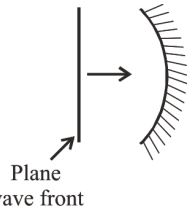
(1) 16

(2) 17

(3) 19

(4) 23

30) When a plane wavefront incident on a concave mirror as shown in figure then shape of reflected wavefront :-



(1)

(2)



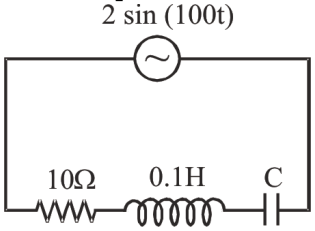
(3)



(4)



31) The power factor of circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to :-



(1) 400 µF

(2) 300 µF

(3) 500 µF

(4) 200 µF

32) Bandwidth of the resonant series L-C-R circuit is :

(1) $\frac{R}{L}$

(2) $\frac{R}{2L}$

(3) $\frac{2R}{L}$

(4) $\frac{4R}{L}$

33) In a double slit experiment, the two slits are 1mm apart and width of each slit is 0.1 mm. A monochromatic light of wavelength 500 nm is used. How many interference fringes of double slit we will get inside the central maximum of single slit pattern ?

(1) 20

(2) 10

(3) 15

(4) 5

34) In parallel plate capacitor the distance between plates is d = 0.1 mm. The medium between the plates is air. The maximum potential difference which can be applied to the capacitor is : (Dielectric strength of air = 3 MV/m)

(1) 3×10^6 volt

(2) 300 volt

(3) 3×10^{10} volt

(4) Infinite

35) Electric potential at any point is V = –5x + 3y + $\sqrt{15}z$. The magnitude of electric field at origin:

(1) $5\sqrt{2}$

(2) zero

(3) 7

(4) $4\sqrt{2}$

36) An electric dipole is placed making at an angle 60° with an electric field of strength 4×10^5 N/C. It experiences a torque equal to $8\sqrt{3}$ N-m. Calculate the charge on the dipole, if it is of length 4 cm.

(1) 10^{-4} C

(2) 10^{-3} C

(3) 10^{-2} C

(4) 10^{-6} C

37) A galvanometer having a resistance of 8 ohm is shunted by a wire of resistance 2 ohm. If the total current is 1 ampere. The part of it passing through the shunt will be :-

(1) 0.25 amp

(2) 0.8 amp

(3) 0.2 amp

(4) 0.5 amp

38) If the distance between plates of a capacitor having capacity C & charge Q is doubled then the work done will be :

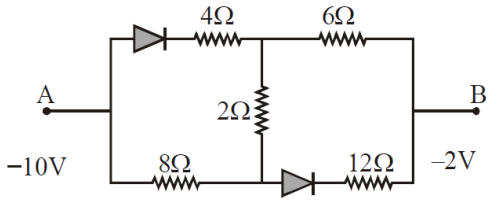
(1) $Q^2/4C$

(2) $Q^2/2C$

(3) Q^2/C

(4) $2Q^2/C$

39) In the following circuit the equivalent resistance between A and B is :-



(1) $\frac{20}{3}\Omega$

(2) 10 Ω

(3) 16 Ω

(4) 20 Ω

40) The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV. The stopping potential in volts is :-

(1) 2

(2) 4

(3) 6

(4) 10

41) Pure Si at 500 K has equal number of electron (n_e) and hole (n_h) concentrations of 1.5×10^{16} m^{-3}. Doping by indium increases n_h to 4.5×10^{22} m^{-3}. The doped semiconductor is of :-

(1) n-type with electron concentration $n_e = 2.5 \times 10^{23}$ m^{-3}

(2) p-type having electron concentration $n_e = 5 \times 10^{9}$ m^{-3}

(3) n-type with electron concentration $n_e = 5 \times 10^{22}\ m^{-3}$

(4) p-type with electron concentration $n_e = 2.5 \times 10^{10}\ m^{-3}$

42) Which statement is correct ?

(1) N-type germanium is negatively charged and P-type germanium is positively charged

(2) Both N-type and P-type germanium are neutral

(3) N-type germanium is positively charged and P-type germanium is negatively charged

(4) Both N-type and P-type germanium are negatively charged

43) In each fission of ${}_{92}U^{235}$, 200 MeV energy is released. How many fissions must occur per second to produce a power of 20 KW :

(1) 3.125×10^{14}

(2) 6.25×10^{14}

(3) 3.125×10^{13}

(4) 6.25×10^{13}

44) If in a photoelectric experiment, the wavelength of incident radiation is reduced from 6000 Å to 4000 Å then :-

(1) Stopping potential will decrease

(2) Stopping potential will increase

(3) Kinetic energy of emitted electrons will decrease

(4) The value of work function will decrease

45) The ratio of de-Broglie wave length of a proton and an α-particle will be 1 : 2 if their :

(1) Kinetic energies are in the ratio 1 : 8

(2) Kinetic energies are in the ratio 8 : 1

(3) Velocities are in the ratio 1 : 8

(4) Velocities are in the ratio 8 : 1

CHEMISTRY

1) What quantity of lime stone on heating will give 56 kg of CaO :-

(1) 1000 kg

(2) 56 kg

(3) 44 kg

(4) 100 kg

2)

In order to increase the rate of the elementary reaction $2A + B \rightarrow$ product, 8 times, it is necessary to :

(1) Make the conc. of A and B three times

(2) Make the conc. of A and B two times

(3) Make the conc. of A and B half

(4) Make the conc. of A and B four times

3) For $2NH_3(g) \rightarrow N_2(g) + 3H_2(g) + 92kCal$, activation energy ($E_a$) for the reaction is 200 kcal. Hence, the activation energy for
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ is :

(1) 108 kcal

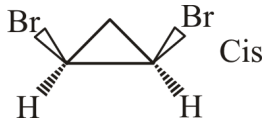
(2) 200 kcal

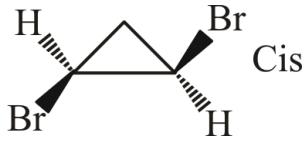
(3) 292 kcal

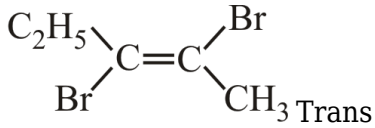
(4) –292 kcal

4) Which of the following compound has incorrect configuration :-

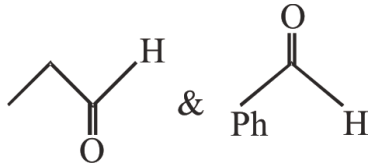
(1) 

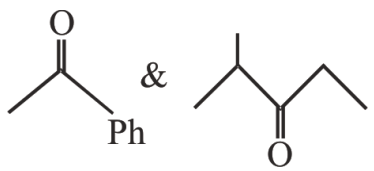
(2) 

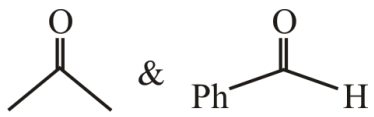
(3) 

(4) 

5) Which of following pair cannot be seperated by using $NaHSO_3$?

(1) 

(2) 

(3) 

(4) All of these

6) Generally bismuth does not form stable pentahalide because of :-

(1) Its higher electronegativity

(2) Its smaller size

(3) Inert pair effect

(4) Non availability of d-orbitals

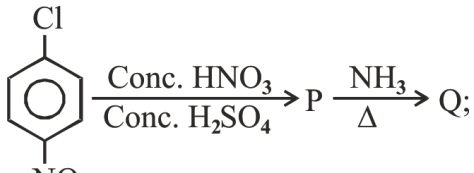
7) The correct statement for the molecule CsI_3 is :-

(1) It contains Cs^{+3} & I^{-3} ions

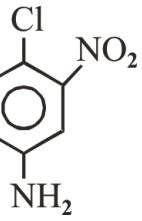
(2) It contains Cs^{+} & I_3^{-} ions

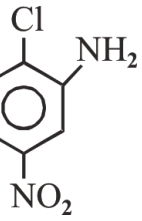
(3) It contains Cs^{+}, I^{-} & I_2 molecule

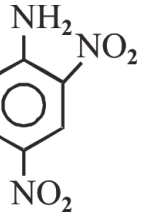
(4) All are correct

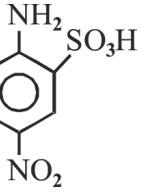


8)
Product Q is:-

(1)


(2)


(3)


(4)


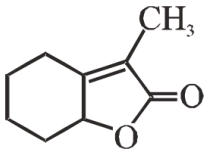
9) For a saturated solution of AgCl at 25°C, k = 1.38×10^{-6} $ohm^{-1}cm^{-1}$. λ°_m for AgCl is 138 $ohm^{-1}cm^2mol^{-1}$, then, the solubility product of AgCl will be

(1) 10^{-5}

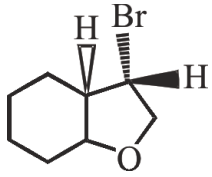
(2) 10^{-10}

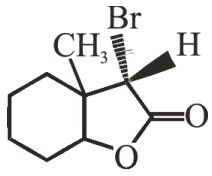
(3) 10^{-14}

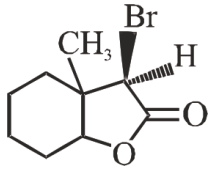
(4) 10^{-16}

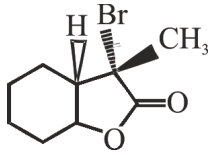
10) 

which of the following reactant is used to obtain above compound [$Et-\overset{\ominus}{O}$ is used during Rxn]

(1) 

(2) 

(3) 

(4) 

11) Calculate the sum of σ and π bonds in the compounds having diagonal hybridisation in the given molecules :

(I) C_6H_6 (II) SO_4^{2-} (III) C_2H_2 (IV) C_2H_4

(1) 5

(2) 4

(3) 15

(4) 6

12) A complex whose IUPAC name is not written correctly is :

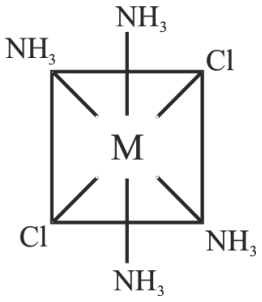
(1) $[Fe(\eta^5\text{-}C_5H_5)_2]$
Bis(η^5–cyclopentadienyl) iron(0)

(2) $[Cr(\eta^6\text{–}C_6H_6)_2]$
Bis(η^6–benzene)chromium(0)

(3) $[CoCl_2(H_2O)_4]Cl.2H_2O$
Tetraaquadichlorocobalt(III) chloride.dihydrate

(4) $[Zn(NCS)_4]^{2-}$
Tetrathiocyanato-N-zincate(II) ion

13) The isomer can be marked as :-



(1) Cis isomer

(2) Trans isomer

(3) d-isomer

(4) □-isomer

14) If 20% nitrogen is present in a compound then its minimum molecular weight will be :-

(1) 144

(2) 28

(3) 100

(4) 70

15) If position and momentum of an electron are determined at a time, then $\Delta P = 3\Delta x$, now the uncertainity in velocity of electron will be :-

(1) $\frac{1}{2m}\sqrt{\frac{h}{3\pi}}$

(2) $\frac{1}{2m}\sqrt{\frac{3h}{\pi}}$

(3) $\frac{1}{4m}\sqrt{\frac{3h}{\pi}}$

(4) $\frac{1}{6m}\sqrt{\frac{h}{\pi}}$

16) $[OH^-]$ in the aqueous solution of equimolar mixture of NH_4OH and NH_4Cl at 25°C is :
(Given K_b for $NH_4OH = 2 \times 10^{-5}$)

(1) 10^{-5} M

(2) 2×10^5 M

(3) 2×10^{-5} M

(4) 10^5 M

17) A first order reaction is carried out with an initial concentration of 10 mol per litre and 60% of the reactant changes into the product. Now if the same reaction is carried out with an initial concentration of 5 mol per litre for the same period the percentage of the reactant changing to the product is.

(1) 40

(2) 80

(3) 160

(4) 60

18) How much oxygen is dissolved in 100 mL water at 298 K if partial pressure of oxygen is 0.9 atm and $K_H = 1.6 \times 10^4$ atm –

(1) 22.4 mg

(2) 1.1 mg

(3) 2.24 g

(4) 0.01 g

19) Number of π e^-s in the Anthracene are :

(1) 7

(2) 14

(3) 9

(4) 16

20) $CH_3–CH_2–OH \xrightarrow[413\ K]{H_2SO_4}$ Major product :-

(1) $CH_2=CH_2$

(2) $CH_3–O–CH_3$

(3) $C_2H_5–O–C_2H_5$

(4) $CH_3–CH=CH–CH_3$

21) Which of the following is not suitable to obtain by Wurtz Reaction?

(1) n – butane

(2) 2,3– Dimethylbutane

(3) n – hexane

(4) 2 – Methylbutane

22) $CH_3–C\equiv C–CH_3 \xrightarrow[Pd + CaCO_3]{H_2} A$, $CH_3–C\equiv C–CH_3 \xrightarrow[Na + liq.\ NH_3]{H_2} B$ Relation between A & B is :-

(1) Chain isomers

(2) Position isomers

(3) Geometrical isomers

(4) Optical isomers

23) Enthalpy change for the reaction

$C_2H_4 + H_2 \rightarrow C_2H_6$ will be :-
Given H–H ⇒ 431.37 kJ/mol
C = C ⇒ 606.10 kJ/mol
C–C ⇒ 336.49 kJ/mol
C–H ⇒ 410.50 kJ/mol

(1) 1523.6 kJ

(2) –243.6 kJ

(3) –120 kJ

(4) 553 kJ

24) Number of optical isomers in the open chain structure of glucose is :

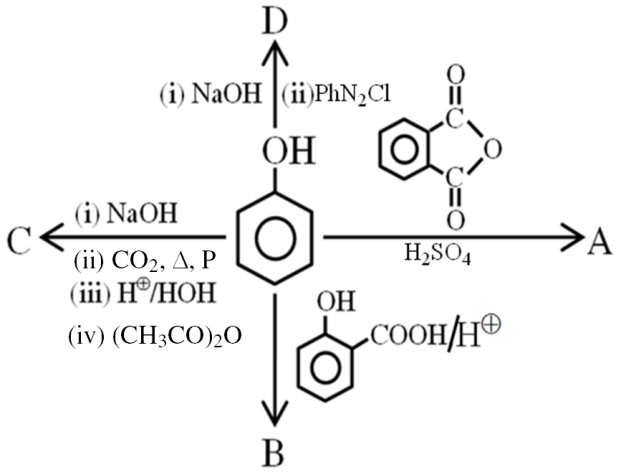
(1) 16

(2) 8

(3) 32

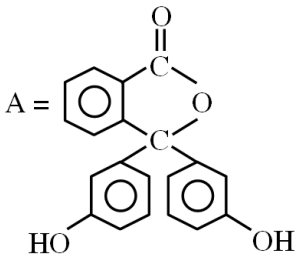
(4) 4

25)



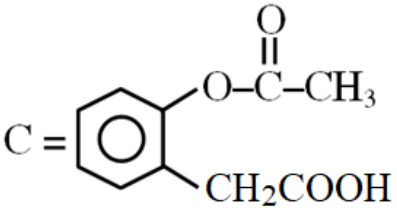
The correct match about above written reaction sequence :

(1)

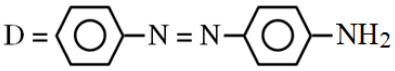


(2) B is phenyl salicylate

(3)



(4)



26) **Statement-I :** Benzene gives Friedal craft Reaction.
Statement-II : All substituted benzene give Friedal craft reaction.
Choose the correct statement :

(1) Statement-I and Statement-II both are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I and Statement-II both are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

27) Match the column :

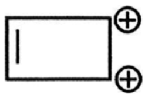
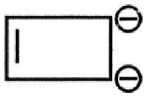
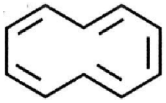
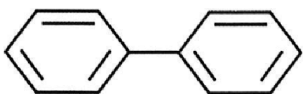
	Column-I		**Column-II**
(A)	$ArN_2^+ Cl^- \rightarrow ArOH$	(p)	$HBF_4/NaNO_2, \Delta$
(B)	$ArN_2^+ Cl^- \rightarrow ArNO_2$	(q)	H_2O
(C)	$ArN_2^+ Cl^- \rightarrow ArH$	(r)	HBF_4, Δ
(D)	$ArN_2^+ Cl^- \rightarrow ArF$	(s)	$CH_3–CH_2–OH$

(1) A - q, B - p, C - s, D - r
(2) A - s, B - p, C - q, D - r
(3) A - q, B - s, C - p, D - r
(4) A - q, B - s, C - r, D - p

28) Which of the following base present in DNA

(1) Thymine
(2) Uracil
(3) Cytosine
(4) Both (1) and (3)

29) Which of the following compound is not aromatic?

(1) 
(2) 
(3) 
(4) 

30) In which of the following 2nd anion is more stable than first ?

(1) $O_2N-\overset{\ominus}{C}H_2$ and $F-\overset{\ominus}{C}H_2$

(2) $\overset{\ominus}{C}F_3$ and $\overset{\ominus}{C}Cl_3$

(3) $F_3C-\overset{\ominus}{C}H_2$ and $Cl_3C-\overset{\ominus}{C}H_2$

(4) $CH_3-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{C}H_2$ and $H_2N-\overset{\ominus}{C}H_2$

31) In the precipitation of the group (III) in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to :

(1) decrease concentration of $OH^{\ominus}$ ions

(2) increase concentration of Cl^- ions

(3) prevent interference by phosphate ions

(4) increase concentration of NH_4^+ ions

32) Match **List-I** with **List-II** :

	List-I **Solid salt treated with dil. H_2SO_4**		**List-II** **Anion detected**
A.	effervescence of colourless gas	I.	NO_2^-
B.	gas with smell of rotten egg	II.	CO_3^{2-}
C.	Yellowish green gas with pungent and suffocating smell	III.	S^{2-}
D.	brown fumes	IV.	Cl^-

Choose the **correct** answer from the options given below :

(1) A-II, B-III, C-IV, D-I

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-I, D-IV

33) When acidified $K_2Cr_2O_7$ solution is added to Sn^{2+} salts then Sn^{2+} changes to :-

(1) Sn

(2) Sn^{3+}

(3) Sn^{4+}

(4) Sn^+

34) The maximum and minimum melting point of first and second transition series elements respectively are obtained with -

(1) Cr and Zn

(2) Cr and Hg

(3) Zn and Mo

(4) Cr and Cd

35)

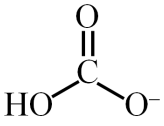
Which of the following property decreases on moving from Ce to Lu ?

(1) Basic nature of oxide

(2) Covalent character of halide

(3) Paramagnetism of trivalent ions

(4) Complex formation tendency

36) Select correct about HCO_3^- :- 

(1) Formal charge on all the O-atom is same

(2) Bond order of all three (C–O) bonds is same.

(3) Bicarbonate ion is nonpolar

(4) All three (C–O) bond lengths are not equivalent.

37) **Assertion :** H_2O & XeF_4 both have plane of symmetry.
Reason : H_2O is polar molecule whereas XeF_4 is non-polar molecule.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

38) Some statement are given below–
(a) Coordination compounds $[Pt(NH_3)_3(NCS)]$ and $[Pt(NH_3)_3(SCN)]$ are examples of linkage isomerism
(b) In the coordination compound $K_4[Ni(CN)_4]$, the oxidation state of nickel is zero.
(c) The hybridization, shape and magnetic moment of $K_3[Cu(CN)_4]$ are respectively dsp^2, square planar, 1.73 B.M.
(d) $[Cr(NCS)(NH_3)_5][ZnCl_4]$ exhibits only Linkage isomerism

(1) only a, b, c are correct

(2) only a, b are correct

(3) only b, c, d are correct

(4) All are correct

39) In which of the following complex species, $\Delta_0 > P$ (pairing energy) with magnetic moment $\sqrt{3}$ B.M.

(1) $[Co(H_2O)_4 (NH_3)_2]^{+3}$

(2) $[Rh F_6]^{3-}$

(3) $[Cr(CO)_6]$

(4) $[Fe(CN)_6]^{3-}$

40) Match the column-I with column-II

Column-I		Column-II	
(A)	$S \rightarrow S^-$	(P)	Endothermic process
(B)	$N^- \rightarrow N$	(Q)	Exothermic process
(C)	$O^- \rightarrow O^{2-}$	(R)	Inert gas configuration in product
(D)	$P \rightarrow P^-$	(S)	Half filled configuration in product

Correct match is :-

(1) A→Q, B→P,R, C→Q,S, D→P

(2) A→Q, B→Q,S, C→Q,R, D→Q,S

(3) A→Q, B→Q,S, C→P,R, D→Q

(4) A→Q,R, B→Q,S, C→P,R, D→P

41) A reaction mixture containing H_2, N_2 and NH_3 has partial pressure 2 atm, 1 atm and 3 atm respectively at 725 K. If the value of K_P for the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$ is 4.28×10^{-5} atm^{-2} at 725 K, in which direction the net reaction will go :

(1) Forward

(2) Backward

(3) No net reaction

(4) Direction of reaction cannot be predicted

42) Calculate the equilibrium constant of the reaction :

$Ni_{(s)} + 2Ag^+_{(aq)} \rightarrow Ni^{+2}_{(aq)} + 2Ag_{(s)}$; $E^o_{cell} = 1.05$ V

(Given : $\frac{2.303RT}{F} = 0.06$)

(1) 10^{35}

(2) 10^{17}

(3) 2×10^{30}

(4) 5×10^{33}

43)

Match the column :

	Column-I		Column-II

(A)	Leclanche cell (Dry cell)	(P)	Converts energy of combustion into electrical energy
(B)	Lead-storage battery	(Q)	Reaction at cathode : $O_2(g)+4H^+(aq)+4e^- \rightarrow 2H_2O(\square)$
(C)	Fuel cell	(R)	Reaction at cathode : $MnO_2+NH_4^+ +e^- \rightarrow MnO(OH)+NH_3$
(D)	Rusting	(S)	Reaction at anode : $Pb(s)+SO_4^{-2}(aq) \rightarrow PbSO_4(s)+2e^-$

(1) A-R, B-S, C-P, D-Q

(2) A-S, B-R, C-P, D-Q

(3) A-P, B-Q, C-R, D-S

(4) A-P, B-R, C-Q, D-S

44) Two liquids A and B are mixed to form an ideal solution. Then match the Column-I with Column-II. (Here all symbol having their usual meaning)

	Column-I		**Column-II**
(A)	$X_A = X_B; P_A^\circ > P_B^\circ$	(I)	$Y_A < Y_B$
(B)	$X_B = 1$	(II)	$P_S = P_B^\circ$
(C)	$X_A < X_B; P_A^\circ = P_B^\circ$	(III)	$Y_A > Y_B$
(D)	$X_A = 1$	(IV)	$P_S = P_A^\circ$

(1) (A–IV), (B–I), (C–III), (D–II)

(2) (A–I), (B–II), (C–III), (D–IV)

(3) (A–II), (B–III), (C–IV), (D–I)

(4) (A–III), (B–II), (C–I), (D–IV)

45) If S_1 be the specific charge (magnitude) of cathode rays and S_2 be that of anode rays, then which is true :

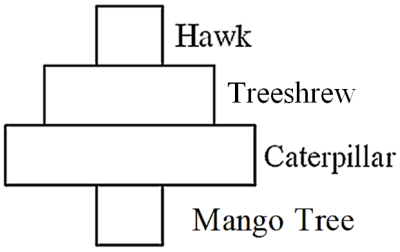
(1) $S_1 = S_2$

(2) $S_1 > S_2$

(3) $S_1 < S_2$

(4) $S_1 \leq S_2$

BIOLOGY

1) The diagram shows a pyramid of number in a food chain - 

What type of organism are caterpillars ?

(1) Primary producer

(2) Primary consumer

(3) Quaternary consumer

(4) Secondary consumer

2) Below two statement are given
Statement - I : Vertical distribution of different species occupying different levels is called stratification.
Statement - II : Productivity is expressed in term of gm^{-2} yr^{-1}.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

3) Hermit crab is the mode of transportation to the Sea-anemone and provide defense system to the Hermit crab, is are example of ?

(1) Co-evolution

(2) Predation

(3) Amensalism

(4) Commensalism

4) Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : The cattle egret bird and grazing cattle in close association. It is classic example of mutualism.
Reason (R) : Plants generally need the help of animals for pollinating their flowers and dispersing their seeds.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true but R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

5) Which one of the following statement is incorrect?

(1) Strict protection of Biodiversity hotspots could reduce ongoing mass extinctions by almost 30%.

(2) 20% of the total oxygen in the earth's atmosphere comes form Amazon rain forest.

(3) The Earth Summit was held in Johannesburg, South Africa on sustainable development.

(4) More than 25% of drugs currently sold in the market world wide are derived from plants.

6) Virus infected cells secrete a protein called __A__ which protect __B__ cells.

(1) A → Interleukins, B → non-infected

(2) A → Interferon, B → infected

(3) A → Interferon, B → non-infected

(4) A → Lymphokines, B → Lymphocytes

7) _____ is the tendency of the body to manifest a characteristic and unpleasant withdrawal syndrome if regular dose of drugs/alcohol is abruptly discontinued.

(1) Euphoria

(2) Adolscence

(3) Dependence

(4) Adaptation

8) All of the following evolved from sauropsids except-

(1) Snakes

(2) Turtles

(3) Tuataras

(4) Mammals

9) **Statement-I :** The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called convergent evolution.
Statement-II : Darwin's finches represent one of the best example of convergent evolution.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Statement II is correct, but statement I is incorrect

(4) Statement II is incorrect, but statement I is correct

10)

Identify the mismatched pair :-

(1) Labeo - Rohu

(2) Trygon - Catla

(3) Torpedo - Electric organ

(4) Scoliodon - Dog fish

11) Stomochord is present in :-

(1) Proboscis region

(2) Collar region

(3) Trunk region

(4) Proboscis gland

12) The partial pressure of oxygen in the alveoli of the lungs is-

(1) Less than that in the blood

(2) Less than that of CO_2 in the blood.

(3) Equal to that in the blood

(4) More than that in the blood

13) Identify the pair of hormones which generate secondary messenger to perform their hormonal action?

(1) Cortisol, Progesterone

(2) Aldosterone, Oestrogen

(3) FSH, Insulin

(4) Progesterone, Testosterone

14) Addison's disease is caused due to underproduction of hormones by -

(1) Adrenal medulla

(2) Adrenal cortex

(3) Anterior pituitary

(4) Thyroid gland

15) Ciliated epithelium are mainly present in the inner surface of hollow organs like _____?

(1) Bronchioles

(2) Fallopian tube

(3) Trachea

(4) All the above

16) Which of the technique is not a part of In Vivo fertilisation :-

(1) G.I.F.T

(2) A.I

(3) I.C.S.I.

(4) Both (1) & (2)

17) The secretion of __(a)__ help the __(b)__ enter into the cytoplasm of the __(c)__ through __(d)__and plasma membrane.
Find the correct option which fills above blanks correctly.

(1)
(a) Acrosome
(b) Ovum
(c) Sperm
(d) Zona pellucida

(2)
(a) Mitochondria
(b) Sperm
(c) Ovum
(d) Zona pellucida

(3)
(a) Acrosome
(b) Sperm
(c) Ovum
(d) Zona pellucida

(4)
(a) Mitochondria
(b) Sperm
(c) Ovum
(d) Zona pellucida

18) The ovum released by the ovary is transported to which region of fallopian tube where fertilisation takes place ?

(1) Isthmus
(2) Ampulla
(3) Infundibulum
(4) Cervix

19) Which of the following contraceptive method is non hormonal ?

(1) LNG-20
(2) Oral contraceptive
(3) Multiload 375
(4) Lactational amenorrhea.

20) Match the column-

Column-I		**Column-II**	
(i)	Fibrinogen	(A)	Involve in defense
(ii)	Globulin	(B)	Help in clotting
(iii)	Albumin	(C)	Help in osmotic balance

Which of the following is correctly match ?
(1) i–A, ii–B, iii–C
(2) i–C, ii–B, iii–A
(3) i–B, ii–A, iii–C
(4) i–B, ii–C, iii–A

21) **Statement-I :-** Axolemma is impermeable to negative charged proteins present in the axoplasm.
Statement-II :- Resting membrane maintained by active-transport of ions by Na^+-K^+ pump which transports $3Na^+$ outwards for $2K^+$ into the cell.

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

22) Muscles which have characteristic striations & involuntary in nature are

(1) Muscles of the eyelids
(2) Muscles assisting locomotion
(3) Muscles of the heart
(4) Muscles in the wall of alimentary canal

23) DNA replication occurs at :-

(1) G_0 & G_1
(2) G_2-stage
(3) S-stage
(4) Metaphase

24) Which of the following is not observed during metaphase?

(1) Position of Centromere
(2) Number of Chromosome
(3) Nucleolus
(4) Size of Chromosome

25) According to fluid mosaic model of cell membrane, the quasi - fluid nature of lipid -

(1) inhibits lateral movement of protein with in overall bilayer.
(2) Describes the arrangements of DNA in the membrane.
(3) Describes the arrangement of ribosome in the membrane.
(4) Enables lateral movement of proteins with in the overall bilayer

26) In cockroach exchange of gases take place at the _____ by diffusion.

(1) Seiracles
(2) Trachea
(3) Tracheoles
(4) Haemocoel

27) ___(i)___ is absent or highly reduced in cortical nephrons.
Fill the blanks (i)

(1) Vasa recta
(2) Collecting duct
(3) Loop of henle

(4) PCT

28) Which of the following factors leads to decrease in population density of a given area at a particular time?

(1) Emigration

(2) Immigration

(3) Mortality

(4) Both 1 and 3

29) Read the following statements regarding ecological pyramid -
i. In most of ecosystem all the pyramids of number, energy & biomass are upright.
ii. The pyramid of biomass in sea is generally upright.
iii. Pyramid of energy is always upright.
iv. Saprophytes are not given any place is ecological pyramids.
How many above statements are incorrect.

(1) (i) only

(2) (ii) only

(3) (ii) & (iii)

(4) All of the above

30) Which of the following statement is incorrect for Ex-situ conservation :

(1) Zoological parks, botanical gardens and wild life safari parks are example of ex-situ.

(2) In this conservation threatened animals and plants are taken out from their natural habitat

(3) Hot spot is example of ex-situ

(4) Seeds of different genetic strains of commercially important plants can be kept for long periods in seed banks

31) Which of the following is well marked in ctenophores ?

(1) Dorso-ventrally flattened

(2) Endoparasites

(3) Triploblastic

(4) Bioluminescence

32) Match the column-I with column-II?

	Column-I		**Column-II**
(A)	Urochordate	(i)	Notochord extend from head to tail
(B)	Cephalochordate	(ii)	Notochord absent
(C)	Vertebrate	(iii)	Notochord in larval tail
(D)	Nonchordata	(iv)	Notochord found during embryonic period.

(1) A–iii, B–i, C–iv, D–ii

(2) A–iv, B–iii, C–i, D–ii

(3) A–i, B–iv, C–iii, D–ii

(4) A–ii, B–iii, C–iv, D–i

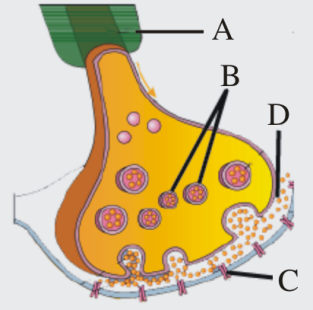
33) How much amount of blood is pumped by each ventricle in 10 cardiac cycles ?

(1) 70 ml

(2) 70 litre

(3) 700 ml

(4) 7 litre

34) Select the correct option: 

(1) A-Synaptic vesicles

(2) B-Post-synaptic membrane

(3) C-Axon

(4) D-Synaptic cleft.

35) The early embryo upto 8 blastomere could than be transferred into fallopian tube the process is known as :

(1) Gamete intra fallopian transfer

(2) Zygote inter fallopian transfer

(3) Gamete inter fallopian transfer

(4) Zygote intra fallopian transfer

36) The space between the plasma membrane of the oocyte and the zona pellucida is called ____. Fill in the blanks ?

(1) Corona radiata

(2) Perivitelline space

(3) Cytoplasm

(4) Germinal vesicle

37) Value of pCO_2 in oxygenated blood is equal to the value of pO_2 in :-

(1) Systemic arteries

(2) Systemic veins

(3) Pulmonary vein
(4) Atmosphere

38) Coca alkaloid or cocaine is obtained from the coca plant _____, native to South America.

(1) *Erythroxylum coca*
(2) *Cannabis sativa*
(3) *Papaver somniferum*
(4) *Atropa belladona*

39) In frog, there are ....A.... pairs of cranial nerve present, what is A

(1) 20-Pairs
(2) 10 Pairs
(3) 5 - pairs
(4) 12-Pairs

40) Which of the following group show all the organised glands?

(1) Pituitary, pineal, heart
(2) Thymus, gonads, adrenal
(3) Parathyroid, kidney, gonads
(4) Pancreas, parathyroid, liver

41)

Match the column I with column II :-

Column I		**Column II**	
(i)	Counter current mechanism	(a)	Check on the Renin angiotensin Mechanism
(ii)	JG Cell	(b)	Powerful vasoconstrictor
(iii)	Angiotensin II	(c)	Release Renin
(iv)	Atrial Natriuretic factor	(d)	Maintain concentration gradient

(1) i-d,ii-c,iii-b,iv-a
(2) i-a,ii-b,iii-c,iv-d
(3) i-b,ii-c,iii-a,iv-d
(4) i-b,ii-a,iii-c,iv-d

42) In white muscles number of ___a___ are less but the amount of ___b___ is high.

(1) a - myoglobin; b - ribosome
(2) a - haemoglobin ; b - proteins
(3) a - mitochondria ; b - myoglobin
(4) a - mitochondria ; b - sarcoplasmic reticulum

43) Which of the following is/are correct with respect to reduction division ?
(A) Diakinesis is marked by appearance of chiasmata
(B) Duplication of centrioles occurs in anaphase-I
(C) Sister chromatids of homologous chromosomes exhibit crossing over
(D) Interkinesis is short lived phase which is followed by prophase-II, a much simpler phase than prophase-I

(1) A, B
(2) B, D
(3) A, C
(4) Only D

44) Which of the following statement is/are correct ?
(a) Cell is the fundamental structural and functional unit of all unicellular organisms only because these organisms are capable of independent existance.
(b) The concentration of ions in vacuole is higher than in the cytoplasm because tonoplast facilitates the transport of ions against concentration gradient into the vacuole.
(c) Cell organelle with cartwheel like organisation has 9 + 0 arrangement of microtubules with 9 peripheral doublets and absence of microtubules in centre.
(d) In submetacentric chromosome, the centromere is situated close to its end forming one extremely short and one very long arm.

(1) a , b , c & d
(2) b, c & d only
(3) b & c only
(4) b only

45) How many hotspots are found in India ?

(1) 25
(2) 34
(3) 5
(4) 3

46) **Assertion :** The content of nucleolus is continuous with the rest of the nucleoplasm.
Reason : Nucleolus is not a membrane bound structure.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

47) How many statements are true ?
(a) Antibodies are proteins in nature.
(b) B-Lymphocytes produce antibodies.
(c) Each antibody molecule has eight peptide chains.
(d) CMI is based on B-cell.

(1) 1

(2) 2

(3) 3

(4) 4

48) During meiosis, separation of homologous chromosomes takes place at :

(1) anaphase-I

(2) diakinesis

(3) metaphase-I

(4) anaphase-II

49) Identify the correct statement :-
(A) Dominance is an autonomous feature of gene.
(B) In co-dominance the F_1 generation can resembles both parents.
(C) Size of starch grain in pea plant is an example of incomplete dominance.
(D) In pleiotropy many genes exhibit single phenotype.
Choose the correct answer from the options given below :

(1) A and D only

(2) B and C only

(3) A, B, C and D

(4) B, C and D

50) Given below are two statements one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Walter Sutton and Theodore Boveri noted that the behaviour of chromosomes was parallel to the behaviour of genes
Reason (R) : Experimental verification of the chromosomal theory of inheritance was done by Thomas Hunt Morgan.

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct and **(R)** is not the correct explanation of **(A)**

(3) **(A)** is are correct but **(R)** is not correct

(4) **(A)** is not correct but **(R)** is correct

51) **Statement-I :** ABO blood groups are controlled by gene I.
Statement-II : The alleles I^A & I^B produce slightly different form of sugar.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct and Statement-II are incorrect.

(3) Statement-I is incorrect and Statement-II are correct.

(4) Both Statement-I and Statement-II are correct.

52) Which of the following RNA polymerase enzyme form hn-RNA :

(1) RNA polymerase-I

(2) RNA polymerase-II

(3) RNA polymerase-III

(4) RNA polymerase I and II

53) Select the incorrect option w.r.t. organisms and their set of chromosomes.

	Organism	**Male**	**Female**
(1)	Human	2A + XY	2A + XX
(2)	Birds	2A + ZW	2A + ZZ
(3)	Grasshopper	2A + XO	2A + XX
(4)	*Drosophila*	2A + XY	2A + XX

(1) 1

(2) 2

(3) 3

(4) 4

54) **Statement-I :** Viruses having RNA genome show shorter life span and evolve faster.
Statement-II : The unequivocal proof that DNA is genetic material came from the experiments of Hershey and Chase.

(1) Both statement I and II are correct

(2) Both statement I and II are incorrect

(3) Statement I is correct and statement II is incorrect

(4) Statement I is incorrect and statement II is correct

55) Find the incorrect statement :

(1) DNA replication takes place in S-phase of the cell cycle of eukaryotes.

(2) Semi-conservative method of DNA replication was first proposed by Watson & Crick.

(3) DNA as an acidic substance present in nucleus was first identified by Franklins.

(4) DNA is a long polymer of deoxyribonucleotides.

56) Which of the following statement is true :-
(A) One turn of the helix in a B-form DNA is approximately 34 nm.
(B) A typical nucleosome contains 200 bps of DNA helix.
(C) In *E.coli* the total length of DNA is 1.36 μm.

(1) B & C only

(2) A & B only

(3) B only

(4) A,B and C

57) Find the correct statement.

(1) m-RNA is called adapter molecule.

(2) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.

(3) Cistron is a segment of DNA coding for a polypeptide.

(4) Aminoacylation of t-RNA is the last phase of translation

58) **Statement-I :** The approach of human genome project which focused on identifying all the genes that are expressed as RNA is referred as sequence annotation.
Statement-II : 99.9 percent of base sequence among human is the same.

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct and Statement-II are incorrect.

(3) Statement-I is incorrect and Statement-II are correct.

(4) Both Statement-I and Statement-II are incorrect.

59) **Assertion :** If one wants to recover many copies of the target DNA , it should be cloned in a vector whose origin support high copy number.
Reason : Origin of replication (ori) is responsible for controlling the copy number of the linked DNA.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

60) The process by which restriction endonuclease cut the DNA molecule :-

(1) Reduction

(2) Oxidation

(3) Hydrolysis

(4) Carboxylation

61)

Given below are two statements :
Statement I : The Indian Government has set up GEAC which will make decisions regarding the validity of GM research.
Statement II : Biopiracy is the term used to refer to the use of bio-resources by indigenous or local people.
In the light of the above statements, choose the correct answer from the options given below :

(1) **Statement** I is true but **Statement II** is false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are true

(4) Both **Statement I** and **Statement II** are false

62) Golden rice is a genetically modified crop plant where the incorporated gene is meant for biosynthesis of :-

(1) Vitamin 'C'

(2) Omega 3
(3) Vitamin 'A'
(4) Vitamin 'B'

63) The first letter of the name of restriction endonuclease came from the :

(1) genus of organism
(2) species of organism
(3) family of organism
(4) class of organism

64) Which of the following organism serve as an important biofertiliser in paddy fields ?

(1) Fungi
(2) Cyanobacteria
(3) Trichoderma
(4) Ladybird

65) Major component of bio gas is :-

(1) Methane
(2) Hydrogen sulphide
(3) Nitrogen oxide
(4) Sulphur oxide

66) Which of the following is not correct about Lichens ?

(1) Lichens are dual organism containing Phycobiont and Mycobiont
(2) They can't be grow in presence of polluted air
(3) They grow very fast in hot and dry region.
(4) They often grow in most inhospitable places such as barren rock.

67) _____ are members of solanaceac family.

(1) Petunia, Tulip
(2) Belladonna, Aloe
(3) Tomato, Brinjal
(4) Tobacco, Lupin

68) Which of the following is first stable product in C4 cycle ?

(1) Oxaloacetic acid
(2) RuBP
(3) Phosphoenol pyruvic acid
(4) 3-phosphoglyceric acid

69) ATP synthesis is linked with _____.

(1) Development of a proton gradient across a membrane.

(2) Development of a electron gradient across a membrane.

(3) Excitation of electrons

(4) Development of Redox potential

70) Match List I with List II :

	List I		**List II**
A	Carboxylation products	i	Required ATP to form RuBP
B	Reduction products	ii	RuBP
C	Regeneration	iii	2 molecules of 3-PGA
D	CO_2 acceptor	iv	Triose phosphate

Choose the correct answer from option given below :

(1) A-iii, B-iv, C-i, D-ii

(2) A-ii, B-iv, C-iii, D-i

(3) A-iv, B-iii, C-i, D-ii

(4) A-i, B-ii, C-iii, D-iv

71) Read the following statements.
Statement – I : Carbon dioxide is the major limiting factor for photosynthesis.
Statement – II : Current availability of CO_2 level is not limiting to the C_3 plants.
Choose the correct answer from the option given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

72) When pericarp is thick and fleshy then it is differentiated into the outer ___A___, the middle ___B___ and the inner ___C___.
Find out the (A), (B) and (C)

Options :	**A**	**B**	**C**
(1)	Endocarp	Mesocarp	Epicarp
(2)	Epicarp	Endocarp	Mesocarp
(3)	Mesocarp	Epicarp	Endocarp
(4)	Epicarp	Mesocarp	Endocarp

(1) 1

(2) 2

(3) 3

(4) 4

73) Match the column-I with column-II and choose the correct option :

Column-I		Column-II	
(i)	Monoecious plant	(a)	Papaya
(ii)	Dioecious plant	(b)	*Hydrilla*
(iii)	Cleistogamous flower	(c)	Castor
(iv)	Water pollination	(d)	*Commelina*

(1) (i) b, (ii) a, (iii) d, (iv) c

(2) (i) c, (ii) a, (iii) d, (iv) b

(3) (i) c, (ii) d, (iii) a, (iv) b

(4) (i) c, (ii) a, (iii) b, (iv) d

74) **Assertion (A) :** The cells of the tapetum are initially diploid and uninueleate but they become polyploid and multinucleate.
Reason (R) : Tapetal cells undergo endomitosis and free nuclear division, respectively.

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** incorrect.

(3) **(A)** is incorrect but **(R)** correct.

(4) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

75) In the polysaccharide the individual monosaccharide are linked by :-

(1) Glycosidic bond

(2) Phosphodiester bond

(3) Hydrogen bond

(4) Peptide bond

76) Classification proposed by Linnaeus is :-

(1) Artificial

(2) Natural

(3) Numerical

(4) Phylogenetic

77) Gas vacuole and heterocyst both are commonly observed in :

(1) *Nostoc*

(2) *Albugo*

(3) Amoeba

(4) Purple photosynthetic bacteria

78) Which statement is incorrect with respect to genus ?

(1) Each genus may have one or more than one specific epithets

(2) It comprises a group of related species

(3) Species of one genera have more characters in common in comparison to species of other genera

(4) Dog and cat belong to same genus

79) How many example in list given below are related with Autotrophic eukaryotes ?
Cycas, Chlorella, Nostoc, Sphagnum, Fucus, Yeast*, Pteris,* Mucor*, Anabaena, Azolla*

(1) 7

(2) 5

(3) 6

(4) 4

80) Find the incorrect statement

(1) Cellular respiration is an amphibolic process

(2) Cellular respiration is an exergonic process.

(3) All the enzymes of TCA cycle are present in mitochondrial matrix.

(4) Yeast poison themselves to death when the concentration of alcohol reaches about 13 percent.

81) Oxidation of one molecule of NADH gives rise to __(a)__ molecules, while that of one molecule of $FADH_2$ produces __(b)__ molecules.

(1) b–3 ATP, a–2ATP

(2) a–2 ATP, b–3ATP

(3) a–3 ATP, b–2ATP

(4) a–4 ATP, b–2ATP

82) Match the column 'A' with column 'B' and select the correct option :

	Column-A		**Column-B**
a	Development	i	First cell of sporophytic generation
b	Zygote	ii	First process of plant growth
c	Seed germination	iii	Fundamental and conspicuous characterstic of a living being
d	Growth	iv	Sum of growth and differentiation

(1) a - iv, b - i, c - ii, d - iii

(2) a - ii, b - iii, c - i, d - iv

(3) a - ii, b - i, c - iv, d - iii

(4) a - iv, b - i, c - iii, d - ii

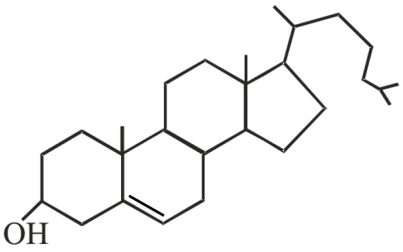
83) Which among the following statement is/are correct?

(a) Glycolysis is also reffered to as EMP pathway.
(b) In glycolysis, glucose undergoes partial oxidation to form two molecules of pyruvic acid.
(c) Citrate is isomerised to isocitrate in TCA cycle.
(d) Complex IV is also known as cytochrome bc_1 complex.

(1) a, b, c
(2) a, b, d
(3) b, c, d
(4) a, c, d

84) The protoxylem lies towards the pith and the metaxylem lies towards the periphery of the plant organ. This type of primary xylem is called _____.

(1) Exarch
(2) Endarch
(3) Mesarch
(4) Centrarch

85) Below structure shows which biomolecule :- 

(1) Adenosine
(2) Alanine
(3) Lecithin
(4) Cholesterol

86) Match the Column-I with Column-II and select correct one?

	Column-I		**Column-II**
(A)	Psilopsida	(i)	*Psilotum*
(B)	Lycopsida	(ii)	*Adiantum*
(C)	Sphenopsida	(iii)	*Selaginella*
(D)	Pteropsida	(iv)	*Equisetum*

(1) (A–i), (B–iii), (C–iv), (D–ii)
(2) (A–i), (B–ii), (C–iii), (D–iv)
(3) (A–i), (B–iii), (C–ii), (D–iv)
(4) (A–i), (B–ii), (C–iv), (D–iii)

87) Agar is obtained from-

(1) Gellidium

(2) Gracillaria

(3) Both (1) & (2)

(4) Chlorella

88) Match list-I with list-II and choose correct option :

	List-I		**List-II**
A	Auxin	(i)	Help overcome the apical dominance
B	Gibberellin	(ii)	Promote flowering in Pineapple
C	Cytokinin	(iii)	Promote bolting in beet
D	Ethylene	(iv)	Synchronising fruit set in pineapple

(1) A-iv, B-i, C-iii, D-ii

(2) A-i, B-ii, C-iii, D-iv

(3) A-ii, B-iii, C-i, D-iv

(4) A-iv, B-i, C-ii, D-iii

89) Which of the following option is incorrect ?

(1) In dicot stems endodermis rich is starch grains so it is also called starch sheath.

(2) In monocotyledonous stems hypodermis made up of parenchyma.

(3) In monocot stem each vascular bundle is surrounded by sclerenchyma.

(4) In dicot stems vascular bundles are arranged in a ring.

90) Match the Column-I and Column-II and choose the correct option.

	Column-I		**Column-II**
(i)	Coralloid roots	(a)	Pinus
(ii)	Mycorrhizal roots	(b)	Wolfia
(iii)	Redwood tree	(c)	Cycas
(iv)	Smallest angiosperm	(d)	Eucalyptus
(v)	Tallest angiosperm	(e)	Sequoia

(1) (i)-(c), (ii)-(e), (iii)-(d), (iv)-(a), (v)-(b)

(2) (i)-(c), (ii)-(a), (iii)-(d), (iv)-(b), (v)-(e)

(3) (i)-(c), (ii)-(a), (iii)-(e), (iv)-(b), (v)-(d)

(4) (i)-(c), (ii)-(a), (iii)-(b), (iv)-(e), (v)-(d)

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	3	1	4	4	3	2	4	2	2	4	2	2	2	4	3	1	1	4	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	2	3	3	2	2	2	3	3	3	1	1	2	3	2	2	2	3	2
Q.	41	42	43	44	45															
A.	2	2	2	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	3	1	4	3	2	3	2	4	1	1	2	4	2	3	4	4	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	3	1	2	2	1	4	3	2	1	1	3	4	1	4	1	2	4	3
Q.	86	87	88	89	90															
A.	2	1	1	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	4	4	4	3	3	3	4	2	2	2	4	3	2	4	3	3	2	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	3	3	3	4	3	1	4	2	3	4	1	3	4	4	2	2	1	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	4	4	4	4	1	2	1	2	2	4	2	2	1	3	3	3	3	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	3	1	2	1	3	3	1	1	1	3	4	2	1	1	1	1	4	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	1	2	4	1	3	3	2	3										

SOLUTIONS

PHYSICS

1) The mean period of oscillation of the pendulum is

$$T_{mean} = \frac{\sum_{i=1}^{n} T_i}{n}$$

$$T_{mean} = \frac{(2.63 + 2.56 + 2.42 + 2.71 + 2.80)}{5} s$$

$$= \frac{13.12}{5} s = 2.624 \text{ s} = 2.62 \text{ s}$$

(Rounded off upto two decimal places)

The absolute errors in the measurement are

$\Delta T_1 = 2.62 \text{ s} - 2.63 \text{ s} = -0.01 \text{ s}$

$\Delta T_2 = 2.62 \text{ s} - 2.56 \text{ s} = 0.06 \text{ s}$

$\Delta T_3 = 2.62 \text{ s} - 2.42 \text{ s} = 0.20 \text{ s}$

$\Delta T_4 = 2.62 \text{ s} - 2.71 \text{ s} = -0.09 \text{ s}$

$\Delta T_5 = 2.62 \text{ s} - 2.80 \text{ s} = -0.18 \text{ s}$

Mean absolute error is

$$T_{mean} = \frac{\sum_{i=1}^{n} |\Delta T_i|}{n}$$

$$\Delta T_{mean} = \frac{(0.01 + 0.06 + 0.20 + 0.09 + 0.18)}{5} s$$

$$= \frac{0.54}{5} s = 0.11 s$$

2)

$v^2 = u^2 + 2aS$

$v^2 = 0^2 + 2aS$

3)

$f_{max} = \mu(mg)$

$= 75 \text{ N}$

4) $\therefore$ $x_1 = x_0$ and $x_2 = 2x_0$

$\therefore$ $$w = \frac{1}{2}K[x_2^2 - x_1^2] = \frac{1}{2}K[(2x_0)^2 - x_0^2] = \frac{3Kx_0^2}{2}$$

$\Rightarrow$ $$\frac{Kx_0^2}{2} = \frac{w}{3}$$

Now $x_1 = 2x_0$ and $x_2 = 3x_0$

$\therefore$ $$w' = \frac{1}{2}K[(3x_0)^2 - (2x_0)^2] = \frac{5Kx_0^2}{2} \Rightarrow \frac{5w}{3}$$

5) $\tau = I\alpha$
or $\tau = (mk^2)\alpha$
$$m = \frac{\tau}{\alpha \times k^2} = \frac{2}{2 \times (2)^2} = \frac{1}{4}kg$$

6)

$Q = \sigma A T^4$
$$T_1 = T + \frac{10}{100}T = 1.1T$$
$Q' = \sigma A (1.1T)^4$
$Q' = \sigma A T^4 (1.46)$
$Q' = 1.46\ \sigma A T^4$
$= \sigma A T^4 + 0.46\sigma A T^4$

46%

7)

By using $v = \sqrt{\frac{3RT}{M}} \Rightarrow u \propto \sqrt{T}$

8)

$A_1 w_1 = A_2 w_2$
$$A_1\sqrt{\frac{k_1}{m}} = A_2\sqrt{\frac{k^2}{m}}$$
$$\frac{A_1}{A_2} = \sqrt{\frac{k_2}{k_1}}$$

9) To produce 5 beats/sec frequency of one wire should be increased upto 505 Hz i.e., increment of 1% in basic frequency
$n \propto \sqrt{T}$ or $T \propto n^2 \Rightarrow \frac{\Delta T}{T} = 2\frac{\Delta n}{n}$
⇒ percentage change in tension = 2(1%) = 2%

10)

Effective gravity on planet
$$g_{eff.} = \frac{G(M/7)}{(R/2)^2} = \frac{4}{9}g_{earth}$$
$(wt.)_P = \frac{4}{9}(wt)_E = 400$ g. wt.

11) Volume flow rate $\frac{dQ}{dt} = \frac{\pi}{8}\frac{pr^4}{\eta\ell}$

$$\Rightarrow \eta = \frac{\pi p r^4}{8\ell} \times \frac{1}{(\text{rate})}$$

$$\eta_1 : \eta_2 : \eta_3 = \frac{1}{2} : \frac{1}{4} : \frac{1}{8} = 4 : 2 : 1$$

12) In reverse case we have to supply heat to melt
= P × t

$$\therefore \quad mL = Pt \Rightarrow L = \frac{Pt}{m}$$

13) Taking horizontal momentum, Initial horizontal momentum before the monkey has jumped
= the final horizontal momentum of the trolley. $(20 + 40) \times 8 = 40 \times V \Rightarrow V = \frac{60 \times 8}{40}$ = 12 m/s.
Initial vertical momentum = 0 when the monkey jumps vertically, it will given a push downwards on the trolley with an equal momentum.

14) $L_i = L_f$

$$\frac{MR^2}{2}\omega = \left[\frac{MR^2}{2} + \frac{M}{2}\left(\frac{R}{2}\right)^2\right]\omega_f$$

$$\frac{MR^2}{2}\omega = \frac{MR^2}{2} \times \frac{5}{4}\omega_f$$

$$\omega_f = \frac{4}{5}\omega$$

15) $e = 1 = \frac{\text{Velocity of separation}}{2V - (-V)}$
Velocity of separation = 3V
Required time = $\frac{2\pi r}{3V}$

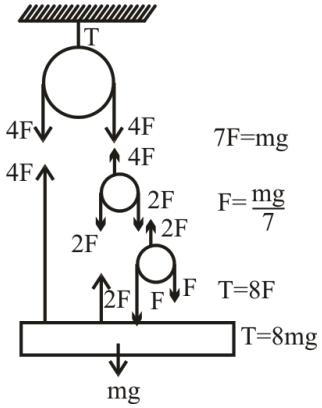
16) In uniform circular motion, magnitude of velocity is constant but direction changes.

17) Pulling force at lowest point = Mg
Pulling force at top most point = 2Mg
Average force $= \frac{Mg + 2Mg}{2} = \frac{3}{2}Mg$

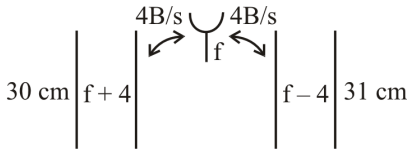
$$y = \frac{\frac{F}{A}}{\frac{\Delta\ell}{\ell}} \Rightarrow \Delta\ell = \frac{F\ell}{Ay}$$

$$\boxed{\Delta\ell = \frac{3Mg\ell}{2yA}}$$

18)



19)



$f \propto \frac{1}{\ell}$

$\frac{f+4}{f-4} = \frac{31}{30}$

$\Rightarrow 30f + 120 = 31f - 124$

$f = 244$ Hz

20) When a charge particle move in electric field such that velocity Electric field and magnetic field all are perpendicular to each other and charge remain undeviated than $\left|\vec{F}_e\right| = \left|\vec{F}_B\right| \Rightarrow qE = qVB \Rightarrow B = \frac{E}{V}$

$\Rightarrow B = \frac{10^4}{10} wb/m^2$

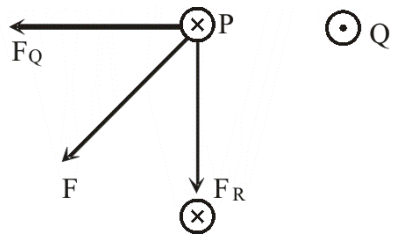
$= 10^3\ wb/m^2$

21) $\frac{1}{v} - \frac{3/2}{+30} = \frac{1-\frac{3}{2}}{+20}$

$\frac{1}{v} = \frac{1}{20} - \frac{1}{40}$

$v = 40$ cm

22) The forces F_Q and F_R are the forces applies by wires Q and R respectively on the wire P as shown in figure. Their resultant force F is best shown by C.



23) $\lambda = \frac{L}{R} = \frac{10}{5} = 2\,\text{sec}$

after t = λ

$I = I_0\left(1 - e^{\frac{-2}{2}}\right)$

$I = I_0 (1 - e^{-1})$

$I^0 = \frac{E}{R} = \frac{5}{5} = 1$

24) $e = NBA\omega; \omega = 2\pi f = 2\pi \times \frac{2000}{60}$

$\therefore e = 50 \times 0.05 \times 80 \times 10^{-4} \times 2\pi \times \frac{2000}{60} = \frac{4\pi}{3}$

25) If cells are connected in series :

$E_{eq} = 2E$, $r_{eq} = 2r$.

Maximum power will be transferred if

$R = r_{eq} = 2r$

Then current in R : $I = \frac{E_{eq}}{R + r_{eq}} = \frac{2E}{2r + 2r} = \frac{E}{2r}$

Power $= I^2R = \left(\frac{E}{2r}\right)^2 2r = \frac{E^2}{2r}$

If cells are connected in parallel ; $E_{eq} = E$, $r_{eq} = \frac{r}{2}$

For maximum power : $R = r_{eq} = \frac{r}{2}$.

Then current in R : $I = \frac{E_{eq}}{R + r_{eq}} = \frac{E}{\frac{r}{2} + \frac{r}{2}} = \frac{E}{r}$

Power $I^2R = \left(\frac{E}{r}\right)^2 \frac{r}{2} = \frac{E^2}{2r}$

Hence [i-q] [ii-q] [iii-s] [iv-r]

26)

If two plane mirrors are kept perpendicular to each other, a ray incident at any angle will always return parallel but in the opposite direction to the incident ray.

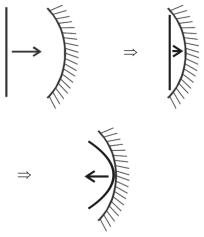
27) $B = \frac{N^2 \mu_0 I \pi}{\ell}$

$\boxed{B \propto N^2}$

28) $\lambda = \frac{2\pi}{K} = \frac{2\pi}{\pi \times 10^3} = 2 \times 10^{-3}$ m

29)

A battery of internal resistance 4Ω
Resistance of network = 4 Ω
$= \frac{(7R)(12R)}{(19R)} = \frac{84R}{19}$
$\Rightarrow R = \frac{19}{21}\Omega$



30)
Reflected wavefront is concave with half radius of curvature.

31) p.f. = $\cos\phi = \frac{1}{\sqrt{2}} = \frac{R}{Z}$
$\Rightarrow Z = \sqrt{2}R \Rightarrow Z = 10\sqrt{2}\Omega$
Here $Z = \sqrt{R^2 + (X_C - X_L)^2}$
where $X_L = = \omega L = (100)(0.1) = 10\ \Omega$
$200 = 100 + (X_C - 10)^2$
$\Rightarrow X_C = 20\Omega$
$\Rightarrow \frac{1}{100C} = 20$
$\Rightarrow C = 500\ \mu F$

32)

β and width $\beta = \frac{R}{L}$

33) Width of CM is diffraction = $\frac{2\lambda D}{a}$
Width of Fringe is YDSE = $\frac{\lambda D}{d}$
If n fringes of YDSE is in the width of CM of single slit diffraction then
$\frac{2\lambda D}{a} = \frac{n\lambda D}{d}$
$\frac{2}{0.1} = \frac{n}{1}$
n = 20

34)

$E = \frac{V}{d}$

Dielectric strength, $E_{max} = \frac{V_{max}}{d}$

$\Rightarrow V_{max} = E_{max}(d)$

$= 3 \times 10^6 \times 0.1 \times 10^{-3}$

$= 3 \times 10^2$ Volt

$= 300$ Volt

35) $\vec{E} = -\frac{dv}{dx}\hat{i} - \frac{dv}{dy}\hat{j} - \frac{dv}{dz}\hat{k}$

$= 5\hat{i} - 3\hat{j} - \sqrt{15}\hat{k}$

$|\vec{E}| = \sqrt{25 + 9 + 15} = 7$ NC^{-1}

36) $\tau = PE \sin\theta = q(d)\, E \sin\theta$

$q = \frac{\tau}{dE\sin\theta} = \frac{8 \times \sqrt{3}}{4 \times 10^{-2} \times 4 \times 10^5 \sin 60} = 10^{-3}C$

37) $I_s = \left(\frac{8}{8+2}\right) \times 1 = 0.8A$

38) $U_{in} = \frac{1}{2}\frac{Q^2}{C}$

On halving the distance $C^1 = \frac{C}{2}$; Q = constant

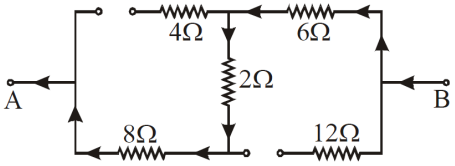
$U_{fin} = \frac{1}{2}\frac{Q^2}{C^1} = \frac{1}{2}\frac{Q^2}{\left(\frac{C}{2}\right)} = \frac{Q^2}{C}$

Hence work done

$W = U_{fin} - U_{in} = \frac{Q^2}{C} - \frac{1}{2}\frac{Q^2}{C} = \frac{Q^2}{2C}$

39) According to the given figure A is at lower potential w.r.t. B. Hence both diodes are in reverse biasing, so equivalent, circuit can be redrawn as follows.

⇒ Equivalent resistance between A and B



R = 8 + 2 + 6 = 16 Ω

40) $K_{max} = (|V_s|)\, eV \Rightarrow |V_s| = 4V$

41) $n_i^2 = n_e n_h$

$(1.5 \times 10^{16})^2 = n_e(4.5 \times 10^{22})$
$n_e = 0.5 \times 10^{10} = 5 \times 10^9$
$n_h = 4.5 \times 10^{22}$
$n_h >> n_e$
Semiconductor is p-type and $n_e = 5 \times 10^9\ m^{-3}$.

42) In N-type semiconductor majority charge carriers are electrons which are negative so N-type. In P-type semiconductor majority charge carriers are holes which are positive so P-type. But overall atom and material is neutral.

43) $\frac{N}{t} = 3.125 \times 10^{10} \times 20 \times 10^3 = 6.25 \times 10^{14}/s$

44)

$\lambda\downarrow$, $K_{man}\uparrow$, $V_0\uparrow$

45) $\lambda = \frac{h}{\sqrt{2mk}}$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha k_\alpha}{m_p k_p}}$$

$$\frac{1}{2} = \sqrt{\frac{4k_\alpha}{k_p}}$$

$$\frac{k_\alpha}{k_p} = \frac{1}{16} \Rightarrow \frac{v_\alpha}{v_p} = \frac{1}{8}$$

CHEMISTRY

46) $CaCO_3 \rightarrow CaO + CO_2$

$$\frac{56 \times 10^3}{56} \text{mol}$$

Moles of $CaCO_3$ = moles of CaO

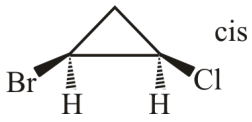
$$= \frac{56 \times 10^3}{56} = 1 \times 10^3 \text{mole}$$

wt of $CaCO_3 = 1 \times 10^3 \times 100$ g
= 100 kg

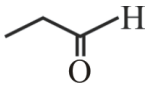
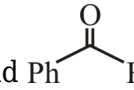
47) Rate = $k[A]^2[B]$
= $k[2A]^2[2B]$
= $8k[A]^2[B]$

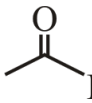
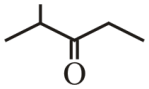
48) $\Delta H = E_f - E_b$
$-92 = 200 - E_b$

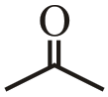
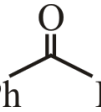
E_b = 292 kcal

49) 

50)

Both  and  react with $NaHSO_3$.

Both  and  does not react with $NaHSO_3$.

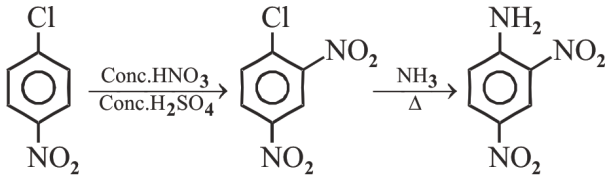
Both  and  react with $NaHSO_3$.

51)

Inert pair effect

52)

It contains Cs^+ & I_3^- ions

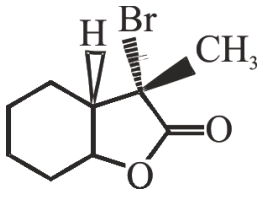
53) 

(P) (Q)

54) $\lambda^0 = \frac{1000 \times K}{S}$

$138 = \frac{1000 \times 1.38 \times 10^{-6}}{S}$

$S = 1 \times 10^{-5}$; $K_{sp} = S^2 = (10^{-10})$

55) 

56)

H–C ≡ C–H (sp hybridisation)

57)

[Fe(η^5 (C_5H_5)$_2$] → Bis(η^5–cyclopentadienyl) iron(II)

58) Trans isomer

59)

70

60) $\frac{1}{2m}\sqrt{\frac{3h}{\pi}}$

61)

Equimolar mixture of NH_4Cl and NH_4OH therefore $[NH_4^+]$: $[NH_4OH]$ = 1 : 1

$pOH = pK_b + \log \frac{[NH_4^+]}{[NH_4OH]}$

or $pOH = 2 \times 10^{-5} + \log \frac{1}{1}$

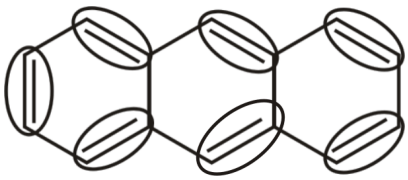
and $[OH^-] = 2 \times 10^{-5}$

62)

60

63)

0.01 g

64) 

Total πe^{Θ} = π bond × 2

= 7 × 2

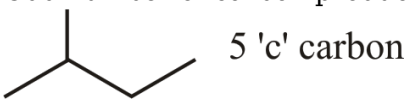
= 14

65)

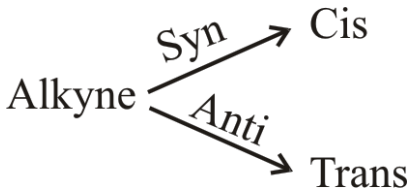
C_2H_5–O–C_2H_5

66)

Odd number of carbon products is very less favoured in wurtz reaction (dimerization or other).



67)



68) $\Delta H = \{606.10+4\times410.5+431.37\}-$
$\{336.49+6\times410.5\} = -120$ kJ/mol

69) No. of optical isomers = $2^n = 2^4 = 16$
n = no. of chiral carbon

70)

B is phenyl salicylate

71)

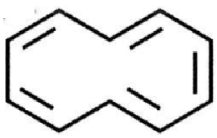
Statement-I is correct but Statement-II is incorrect.

72)

A - q, B - p, C - s, D - r

73)

Both (1) and (3)

74) 

75) $\overset{\ominus}{C}F_3$ and $\overset{\ominus}{C}Cl_3$

76)

decrease concentration of $OH^{\ominus}$ ions

77)

Solid salt treated with dil. H_2SO_4	Anions detected
(A) effervescence of colourless gas →	CO_3^{-2}
(B) gas with smell of rotten egg →	S^{2-}
(C) Yellowish green gas with pungent and suffocating smell →	Cl^-
(D) Brown fumes →	NO_2^-

78)

Sn^{4+}

79)

3d series Max M.P = Cr
4d series Min M.P = Cd

80) Ce → Lu
Z_{eff} ↑ size ↓ Polarisation ↑ CoV. ↑ Acidic ↑ Basic ↓

81)

All three (C–O) bond lengths are not equivalent.

82)

Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

83)

only a, b are correct

84)

$[Fe(CN)_6]^{3-}$

85)

A→Q, B→Q,S, C→P,R, D→Q

86)

Backward

87) NCERT Pg # 40, Part-I

88) NCERT Pg#86 to 89

89)

Raoult's law:
$P_A = X_A P_A^\circ$, $P_B = X_B P_B^\circ$, $P = X_A P_A^\circ + X_B P_B^\circ$
$Y_A = P_A / (P_A + P_B) = (X_A P_A^\circ)/(X_A P_A^\circ + X_B P_B^\circ)$
$Y_B = P_B / (P_A + P_B) = (X_B P_B^\circ)/(X_A P_A^\circ + X_B P_B^\circ)$
(A) $X_A = X_B; P_A^o > P_B^o$, A is more volatile ⇒ $Y_A > Y_B$ ⇒ matches (III).
(B) $X_B = 1$, Only B present ⇒ solution pressure (P)= P_B^o ⇒ matches (II).
(C) $X_A < X_B; P_A^o = P_B^o$, Equal pure vapour pressures → vapour phase composition follows liquid composition ⇒ $Y_A < Y_B$ matches (I).
(D) $X_A = 1$, Only A present total pressure (P) = P_A^o ⇒ matches (IV).
Final answer: Option (4)

90)

Specific charge = $\frac{\text{charge}}{\text{mass}}$

BIOLOGY

91) NCERT XII Pg No. # 212

92)

NCERT-XII, Pg. No. # 207

93) NCERT XII Pg. # 197

94) NCERT-XII Pg. # 201, 202

95)

NCERT XII, Pg. 222, 225

96)

A → Interferon, B → non-infected

97) NCERT Pg, 160

98) NCERT Page No. 123

99) NCERT XII Pg # 117

100) NCERT Pg. # 48

101) NCERT Pg. # 46

102)

Module-6 Page No.#19

103)

FSH, Insulin

104) NCERT Pg. # 244

105) NCERT Pg. No. # 101

106) NCERT Pg. # 48

107) NCERT Pg # 35-36

108) NCERT, Pg. # 35

109) **Question Explanation –** The question asks which contraceptive method is non-hormonal.
Concept – Non-hormonal method
Solution – Multiload 375 is a copper device and does not use hormones.
Final Answer – Option 3

110) NCERT Pg. # 193

111) NCERT-XI Pg. # 317

112) NCERT-XI Pg#304, Ist para

113)

S-stage

114)

Nucleolus

115)

Enables lateral movement of proteins with in the overall bilayer

116) NCERT Pg. # 114

117) NCERT Pg. # 208

118) NCERT Pg. # 193

119) NCERT Pg.# 213

120) NCERT, Pg # 225

121) NCERT, Pg. # 42

122)

A–iii, B–i, C–iv, D–ii

123) NCERT Pg. # 200

124) NCERT-XI Page No. 234

125) NCERT Pg # 48

126) NCERT, Pg. # 35

127)

Question Explanation: The question asks to identify which site has a PO_2 equal to the PCO_2

in oxygenated blood.
Concept: Respiratory physiology – Partial pressures of gases
Solution:

- Oxygenated blood (systemic arteries/pulmonary veins) has $PCO_2 \approx 40$ mm Hg.
- PO_2 in systemic veins (deoxygenated blood returning to heart) ≈ 40 mm Hg, which matches the PCO_2 in oxygenated blood.

Final Answer: Option 2

128) *Erythroxylum coca*

129) NCERT Pg. # 18

130)

Question Explanation –
The question asks which group includes only organised glands, i.e., glands that are structurally well-defined and secrete hormones.

Concept – Endocrine glands

Solution –

A. Organised glands are well-defined structures that secrete hormones directly into blood.

B. Thymus, gonads, and adrenal glands are all organised endocrine glands.

C. Heart, kidney, and liver are organs that secrete hormones but are not considered organised endocrine glands.

So, the group containing only organised glands is Thymus, gonads, adrenal.

Final Answer – Option 2

131) NCERT Pg. # 211,212,213

132) NCERT Pg. No. # 222-223

133)

Only D

134)

NCERT XI Pg # 134

135) NCERT, Pg # 224

136) NCERT Pg#138

137) NCERT Pg # 135 (E), 163 (H)

138)

139) NCERT XII Pg. # 60,61,69

140) NCERT-XII, Pg. # 67

141) NCERT-XII, Pg. # 61, 62

142) NCERT-XII, Pg # 95

143) NCERT XII Page No. # 70

144) NCERT-XII Pg # 86

145) NCERT-XII, Page No. # 88,89

146) NCERT-XII, Page No. # 83,85

147) NCERT-XII, Page No. # 94,95

148) NCERT-XII, Pg. # 103,105

149) NCERT XII Pg # 199

150)

NCERT-XII, Pg. # 196

151)

NCERT-XII, Pg. # 184,185

152)

Vitamin 'A'

153)

The first letter of the name of a restriction endonuclease typically comes from the genus of the organism from which it was isolated. For example, EcoRl is derived from Escherichia coll (genus Escherichia), with "Eco" coming from the genus name.

154)

Explain Question - The question asks which organism acts as an important biofertiliser in paddy fields.

Concept - Microbes As Biofertilizers

Solution -
Fungi: not the main biofertiliser in paddy fields.
Cyanobacteria: fix nitrogen, widely used in rice cultivation → correct.
Trichoderma: used as biocontrol agent, not biofertiliser.
Ladybird: insect, not a biofertiliser.

Final Answer - Option 2

155) NCERT (XIIth) Pg. # 185

156) NCERT Pg. No. # 27 & Module

157)

NCERT Pg. No. 68

158) NCERT-XI, Pg # 146

159) NCERT, Pg. # 141

160)

NCERT-XI, Pg # 144

161) NCERT XI, Pg. # 150

162) NCERT-XI, Pg. 65

163) NCERT-XII Pg. 12,13,14

164) NCERT XII Pg. # 6

165)

Explanation:

A. In **polysaccharides**, individual **monosaccharide units** are linked together by **glycosidic bonds**.

B. A **glycosidic bond** is a type of **covalent bond** formed between the hydroxyl group of one sugar and the anomeric carbon of another.

Option 1: Glycosidic bond

166)

Module-1

167)

NCERT-XI, Pg. # 19, 129

168)

NCERT XI Pg.# 9

169) NCERT Pg. No. # 223

170)

NCERT XI, Page No. - 227,230 + Module

171)

NCERT XI Page No. 233

172) NCERT Pg. No. # 239

173) NCERT XI Pg. # 228, 231, 233

174) NCERT Page No. 87, 90

175) NCERT XI, Pg.# 107

176) NCERT Pg # 32

177) NCERT Pg # 23-25

178) NCERT Pg. No. # 177

179) NCERT Pg. # 76

180)

NCERT XI - P. No. - 38,40